

Model selection

- Problem setup
- Linear regression

Regression problems

We have input/output data

$$\mathcal{D} = \left\{ (x_n, y_n) \right\}_{n=1}^N \subset \mathbb{X} \times \mathbb{Y}$$

and seek a **regression function**

$$f : \mathbb{X} \mapsto \mathbb{Y}$$

However, we are **unsure** about the basis

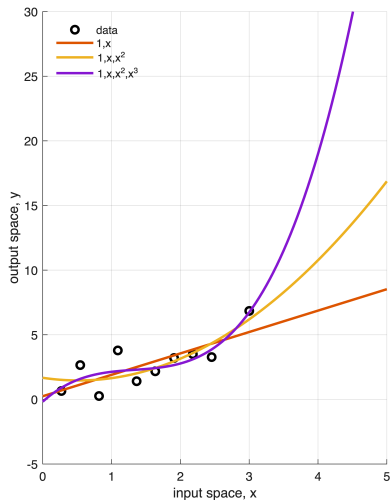
$$\phi^m : \mathbb{X} \mapsto \mathbb{Y}, \quad m = 1, \dots, M$$

that forms the **feature expansion**

$$\phi(x) = [\phi^1(x), \dots, \phi^M(x)]$$

of our **linear regression model**

$$f(x) = \phi(x)c = \sum_{m=1}^M c^m \phi^m(x)$$



Model selection problems

In general, we have a **pool** with K different models

The k^{th} **model** in the pool has **size** M_k ...and its **basis** consists of

$$\phi_k^m : \mathbb{X} \mapsto \mathbb{Y}, \quad m = 1, \dots, M_k$$

...and its **feature expansion** is

$$\phi_k(x) = [\phi_k^1(x), \dots, \phi_k^{M_k}(x)]$$

...and its **linear regression model** is

$$f(x) = \phi_k(x)c_k = \sum_{m=1}^{M_k} c_k^m \phi_k^m(x)$$

...and its **unknown coefficients** are

$$c_k = \begin{pmatrix} c_k^1 \\ \vdots \\ c_k^{M_k} \end{pmatrix} \in \mathbb{R}^{M_k}$$

The models in our pool

- may or may not be **nested**
- may or may not have same **size**
- may or may not share **features**

The only strict requirements is that all feature functions are **compatible** with our input and output data-spaces

$$\begin{aligned} \phi_k^m : \mathbb{X} &\mapsto \mathbb{Y} \\ m &= 1, \dots, M_k \\ k &= 1, \dots, K \end{aligned}$$

Model selection asks for **choosing one** of the models in the pool irrespective of their optimal coefficients

Example

Our pool contains $K = 3$ models

- model $k = 1$:

$$M_1 = 2 \quad \phi_1(x) = [1, x]$$

$$c_1 \in \mathbb{R}^2 \quad f(x) = \phi_1(x)c_1 = c_1^1 + c_1^2 x$$

- model $k = 2$:

$$M_2 = 3 \quad \phi_2(x) = [1, x, x^2]$$

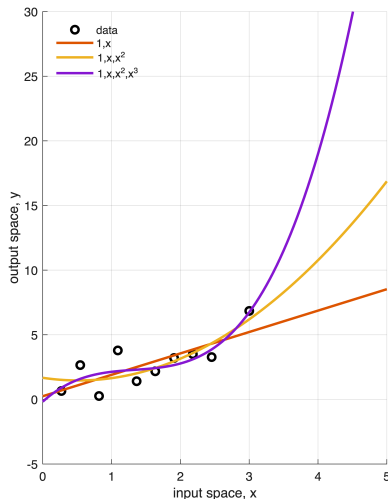
$$c_2 \in \mathbb{R}^3 \quad f(x) = \phi_2(x)c_2 = c_2^1 + c_2^2 x + c_2^3 x^2$$

- model $k = 3$:

$$M_3 = 4 \quad \phi_3(x) = [1, x, x^2, x^3]$$

$$c_3 \in \mathbb{R}^4 \quad f(x) = \phi_3(x)c_3 = c_3^1 + c_3^2 x + c_3^3 x^2 + c_3^4 x^3$$

The models in our pool are **nested**



Bayesian model selection

We have a **pool** with K models, each characterized by potentially different sizes $M_{1:K}$, feature expansions $\phi_{1:K}(x)$, and coefficients $c_{1:K}$

Each model carries its own **Bayesian formulation**

$$\begin{aligned} c_k &\sim \text{Normal}_{M_k}(\mu_k, C_k), & k = 1, \dots, K \\ y_n | c_k &\sim \text{Normal}(\phi_k(x_n) c_k, \zeta), & n = 1, \dots, N, \quad k = 1, \dots, K \end{aligned}$$

with model-specific **prior hyper-parameters** $\mu_k \in \mathbb{R}^{M_k}$ and $C_k \in \mathbb{R}_{\text{sym},+}^{M_k \times M_k}$

Combining all models in the pool, we obtain a **master Bayesian formulation**

$$\begin{aligned} \kappa &\sim \text{Categorical}_{1:K}(\pi_1, \dots, \pi_K) \\ c_k &\sim \text{Normal}_{M_k}(\mu_k, C_k), & k = 1, \dots, K \\ y_n | \kappa, c_{1:K} &\sim \text{Normal}(\phi_\kappa(x_n) c_\kappa, \zeta), & n = 1, \dots, N \end{aligned}$$

This way **model selection** reduces to **hypothesis testing**

...with uncertain parameters

Bayesian model selection

We have a **pool** with K models, each characterized by potentially different sizes $M_{1:K}$, feature expansions $\phi_{1:K}(x)$, and coefficients $c_{1:K}$

Our **master Bayesian formulation** consists of

$$\begin{aligned}\kappa &\sim \text{Categorical}_{1:K}(\pi_1, \dots, \pi_K) \\ c_k &\sim \text{Normal}_{M_k}(\mu_k, C_k), & k = 1, \dots, K \\ y_n | \kappa, c_{1:K} &\sim \text{Normal}(\phi_\kappa(x_n) c_\kappa, \zeta), & n = 1, \dots, N\end{aligned}$$

We select a model according to the **MAP criterion**

$$\kappa_* = \operatorname{argmax}_{k=1, \dots, K} p(\kappa | y_{1:N})$$

which is based on the **marginal posterior**

$$p(\kappa | y_{1:N}) = \int dc_1 \cdots \int dc_K p(\kappa, c_{1:K} | y_{1:N})$$

Bayesian model selection

Our master Bayesian formulation consists of

$$\begin{aligned}\kappa &\sim \text{Categorical}_{1:K}(\pi_1, \dots, \pi_K) \\ c_k &\sim \text{Normal}_{M_k}(\mu_k, C_k), \\ y_n | \kappa, c_{1:K} &\sim \text{Normal}(\phi_\kappa(x_n) c_\kappa, \zeta),\end{aligned}\quad \begin{aligned} & \\ & \\ n = 1, \dots, N \end{aligned} \quad k = 1, \dots, K$$

which is the same as

$$\begin{aligned}\kappa &\sim \text{Categorical}_{1:K}(\pi_1, \dots, \pi_K) \\ c_k &\sim \text{Normal}_{M_k}(\mu_k, C_k), \\ Y | \kappa, c_{1:K} &\sim \text{Normal}_N(\Phi_\kappa c_\kappa, \zeta I)\end{aligned}\quad k = 1, \dots, K$$

where, as usual, we gather data and features in

$$Y = \begin{pmatrix} y_1, \\ \vdots \\ y_N \end{pmatrix}, \quad \Phi_k = \begin{bmatrix} \phi_k(x_1) \\ \vdots \\ \phi_k(x_N) \end{bmatrix} = \begin{bmatrix} \phi_k^1(x_1) & \cdots & \phi_k^{M_k}(x_1) \\ \vdots & \ddots & \vdots \\ \phi_k^1(x_N) & \cdots & \phi_k^{M_k}(x_N) \end{bmatrix}, \quad k = 1, \dots, K$$

Bayesian model selection

Our **master Bayesian formulation** consists of

$$\begin{aligned}\kappa &\sim \text{Categorical}_{1:K}(\pi_1, \dots, \pi_K) \\ c_k &\sim \text{Normal}_{M_k}(\mu_k, C_k), \\ Y|\kappa, c_{1:K} &\sim \text{Normal}_N(\Phi_\kappa c_\kappa, \zeta I)\end{aligned}\quad k = 1, \dots, K$$

Our **marginal posterior** is given by

$$\begin{aligned}p(\kappa|Y) &= \int dc_1 \cdots \int dc_K p(\kappa, c_{1:K}|Y) \\ &\propto \int dc_1 \cdots \int dc_K p(Y|\kappa, c_{1:K}) p(\kappa, c_{1:K}) \\ &= \int dc_1 \cdots \int dc_K p(Y|\kappa, c_{1:K}) p(\kappa) p(c_{1:K}) \\ &= p(\kappa) \int dc_1 \cdots \int dc_K p(Y|\kappa, c_{1:K}) \prod_{k=1}^K p(c_k) \\ &= p(\kappa) \int dc_1 \cdots \int dc_K p(Y|c_\kappa) \prod_{k=1}^K p(c_k) = p(\kappa) \int dc_\kappa p(Y|c_\kappa) p(c_\kappa)\end{aligned}$$

Bayesian model selection

Our **master Bayesian formulation** consists of

$$\begin{aligned}\kappa &\sim \text{Categorical}_{1:K}(\pi_1, \dots, \pi_K) \\ c_k &\sim \text{Normal}_{M_k}(\mu_k, C_k), \\ Y|\kappa, c_{1:K} &\sim \text{Normal}_N(\Phi_\kappa c_\kappa, \zeta I)\end{aligned}\quad k = 1, \dots, K$$

Our **marginal posterior** is given by

$$\begin{aligned}p(\kappa|Y) &\propto p(\kappa) \int dc_\kappa p(Y|c_\kappa) p(c_\kappa) \\ &= \pi_\kappa \int dc_\kappa \text{Normal}_N(Y; \Phi_\kappa c_\kappa, \zeta I) \text{Normal}_{M_\kappa}(c_\kappa; \mu_\kappa, C_\kappa) \\ &= \pi_\kappa \text{Normal}_N(Y; \mu'_\kappa, C'_\kappa)\end{aligned}$$

where the **updated hyper-parameters** are given by

$$\begin{aligned}\mu'_\kappa &= \Phi_\kappa \mu_\kappa \\ C'_\kappa &= \Phi_\kappa C_\kappa \Phi_\kappa^t + \zeta I\end{aligned}$$

Example

With a pool of $K = 3$ models and sizes

$$M_1 = 2, \quad M_2 = 3, \quad M_3 = 4$$

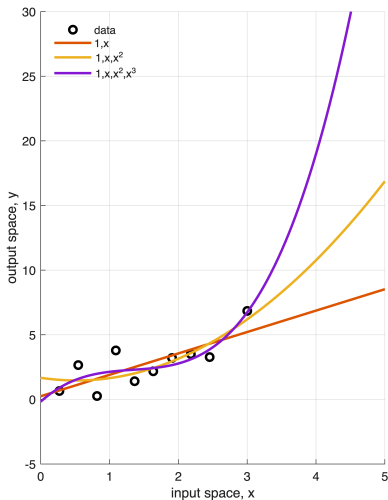
Our master Bayesian formulation consists of priors

$$\kappa \sim \text{Categorical}_{1:3} \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right)$$

$$c_1 \sim \text{Normal}_2 \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{bmatrix} 1, 0 \\ 0, 1 \end{bmatrix} \right)$$

$$c_2 \sim \text{Normal}_3 \left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{bmatrix} 1, 0, 0 \\ 0, 1, 0 \\ 0, 0, 1 \end{bmatrix} \right)$$

$$c_3 \sim \text{Normal}_4 \left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{bmatrix} 1, 0, 0, 0 \\ 0, 1, 0, 0 \\ 0, 0, 1, 0 \\ 0, 0, 0, 1 \end{bmatrix} \right)$$



Example

With a pool of $K = 3$ models and sizes

$$M_1 = 2, \quad M_2 = 3, \quad M_3 = 4$$

Our master Bayesian formulation consists of a likelihood

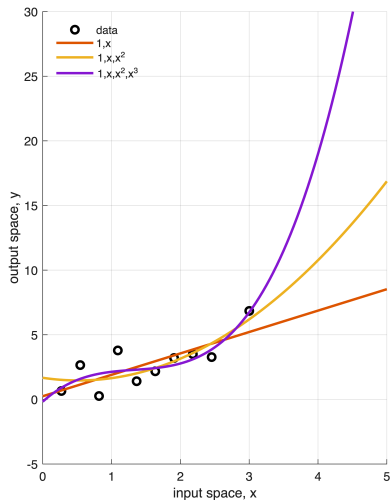
$$Y|\kappa, c_{1:3} \sim \text{Normal}_N(\Phi_\kappa c_\kappa, I)$$

with designs

$$\Phi_1 \in \mathbb{R}^{N \times 2}$$

$$\Phi_2 \in \mathbb{R}^{N \times 3}$$

$$\Phi_3 \in \mathbb{R}^{N \times 4}$$



Example

With a pool of $K = 3$ models and sizes

$$M_1 = 2, \quad M_2 = 3, \quad M_3 = 4$$

and $N = 10$ datapoints, our **marginal posterior** evaluates at

$$p(\kappa = 1|Y) \propto 0.14 \times 10^{-8}$$

$$p(\kappa = 2|Y) \propto 0.23 \times 10^{-8}$$

$$p(\kappa = 3|Y) \propto 0.06 \times 10^{-8}$$

indicating that **model 2** is selected

This model has a feature expansion of

$$\phi(x) = [1, x, x^2]$$

